

PREPRINT

## Exploring Equivalence Testing with the Updated TOSTER R Package

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### ARTICLE HISTORY

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### ABSTRACT

Despite decades of methodological guidance, researchers continue to misinterpret non-significant results as evidence for the absence of an effect. Equivalence testing via two one-sided tests (TOST) provides a principled alternative, but adoption remains limited by software accessibility and lack of training. The TOSTER R package, originally developed by Lakens (2017), made TOST accessible to experimental psychologists. This paper describes a major update to the package that expands its scope with (1) a unified t-test interface supporting both equivalence and minimal effect testing, (2) a simplified hypothesis testing interface returning standard R objects, (3) tools for computing and testing standardized effect sizes and correlations, (4) robust and resampling-based alternatives including bootstrap and permutation tests, (5) equivalence testing for ANOVAs, and (6) power analysis and sample size planning. Each capability is demonstrated with worked examples. TOSTER is also available as a jamovi module for researchers who prefer a graphical interface.

### KEYWORDS

statistics, bootstrap, minimal effects test, NHST, TOST, R package, robust statistics

## 1. Introduction

Researchers routinely interpret non-significant p-values from a nil-hypothesis (i.e., null hypothesis that the mean difference equals zero) test as evidence that no effect exists — a logical error that has been recognized for decades (Altman and Bland 1995). A non-significant result means only that the data are compatible with the null hypothesis; it does not mean the null hypothesis is true.

Equivalence testing provides a principled framework for evaluating whether an effect is small enough to be considered negligible. The two one-sided tests (TOST) procedure is the most widely used approach (Schuirmann 1987). Researchers specify an upper ( $\Delta_U$ ) and lower ( $\Delta_L$ ) equivalence bound based on the smallest effect size of interest (SESOI). Two one-sided tests then evaluate whether the observed effect falls within those bounds. If both tests reject at a pre-specified alpha level, the effect can be considered practically equivalent to zero (Lakens 2017).

The TOSTER R package<sup>1</sup> was originally developed by Lakens (2017) to introduce

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<sup>1</sup>All updates to the package can be found on the package's website <https://aaroncaldwell.us/TOSTERpkg>

experimental psychologists to the concept of equivalence testing and provide an accessible implementation in R. The package provided dedicated functions for independent, paired, and one-sample t-tests, as well as correlations and meta-analytic equivalence tests. It has been widely adopted, with over 1000 citations to date.

However, the original package had a number of limitations. It offered a function-per-design interface (separate functions for each t-test type), lacked robust alternatives to parametric tests, provided no support for ANOVA equivalence testing, and had no tools for comparing effect sizes across studies. The package also did not integrate with standard R workflows and usually just printed results to the console rather than returning objects that could be manipulated or plotted.

This paper provides a tutorial-style introduction to the updated TOSTER package, organized around its major capabilities: (1) a unified t-test interface (`t.TOST`) that pairs a standard significance test with an equivalence test and supports minimal effect testing, (2) a simplified hypothesis testing interface (`simple_htest`) that returns standard `htest` objects, (3) tools for computing and testing standardized effect sizes — including standardized mean differences, rank-based effect sizes, and correlations, (4) robust and resampling-based alternatives including the Brunner-Munzel test, bootstrap and permutation t-tests, and log-transformed methods, (5) equivalence testing for ANOVAs, and (6) power analysis and sample size planning. Each section motivates when and why a researcher would use the tool before demonstrating how.

TOSTER occupies a specific niche in the R ecosystem in that it is a frequentist toolkit for equivalence testing of standard designs. Researchers interested in Bayesian approaches to evidence for the null have excellent options (e.g., `bayestestR`). For equivalence-style contrasts in more complex models — mixed models, marginal means — the `emmeans` and `marginalEffects` packages can accommodate non-zero null hypotheses. TOSTER fills the space of accessible, self-contained equivalence testing for common, albeit simple, research designs. For a broader introduction to the methodology, see [Lakens \(2017\)](#), [Lakens, Scheel, and Isager \(2018\)](#), [Lakens et al. \(2020\)](#), and [Mazzolari et al. \(2022\)](#). TOSTER is also available as a `jamovi` module for researchers who prefer a graphical interface although it has limited capabilities compared to the R package.

## 2. Setting Equivalence Bounds

Before applying any equivalence test, the researcher must specify the equivalence bounds which defines the range of effect sizes considered too small to be meaningful. There is no single correct method for choosing these bounds; the appropriate approach depends entirely on the research question and domain of research.

If a researcher has an equivalence hypothesis, they should be able to articulate what magnitude of effect would be too small to matter in context. The SESOI must flow from the hypothesis formulated by theory and context not from a formula. However, there are several general strategies that have been evaluated ([Cook et al. 2018](#)). Anchor-based methods tie the bound to an externally meaningful criterion, such as a clinically meaningful difference or a “just-noticeable” difference ([Anvari and Lakens 2021](#)). Distribution-based methods define the bound as a fraction of the measurement error ([Jacobson and Truax 1991](#)), though this approach is generally not recommended ([Cook et al. 2018](#)). Cost-benefit, also referred to as health economic methods ([Cook et al. 2018](#)), reasoning sets the bound at the point where there is a net benefit relative to the cost of an intervention. Researchers can draw on existing literature or norma-

tive data to identify effect magnitudes that prior work suggests are negligible. When theory is sufficiently precise, however, the bound may require no external benchmark at all, and the hypothesis itself implies what magnitude of effect would be too small to matter.

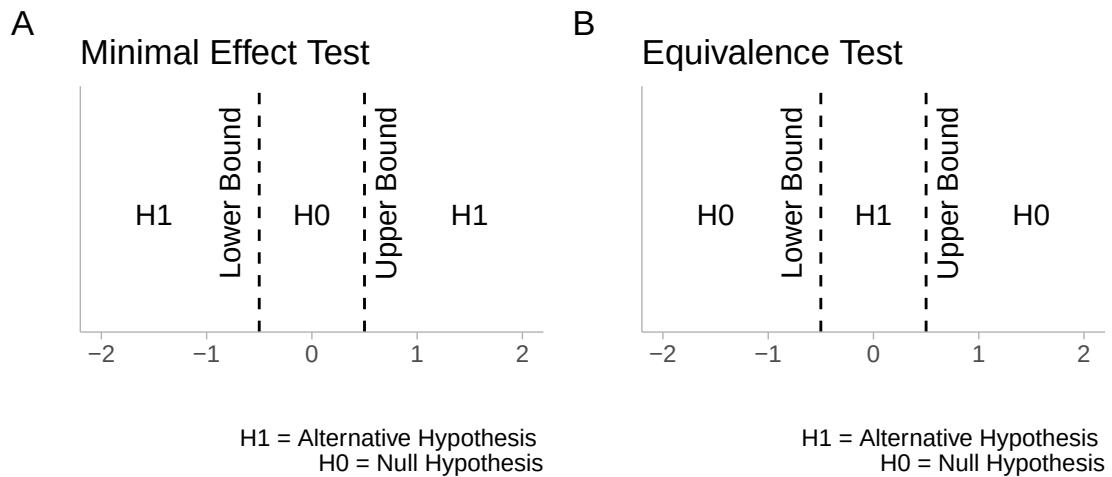
Using “default” bounds on the standardized mean difference (SMD; also known as Cohen’s  $d$ ) based on Cohen’s benchmarks (e.g.,  $SMD = 0.2, 0.5, 0.8$  to imply small, medium, or large effects, respectively) without domain-specific justification is strongly discouraged. Cohen himself stated that these conventions were “recommended for use only when no better basis for estimating the index is available” (p. 25, as cited in [Panzarella, Beribisky, and Cribbie \(2021\)](#)) and reportedly came to regret having proposed them at all ([Funder and Ozer 2019](#)). These benchmarks were never intended as universal thresholds; applied uncritically, they can produce equivalence bounds that are too wide, too narrow, and likely entirely disconnected from the substantive research question.

If a researcher cannot specify meaningful bounds, this itself is informative — it suggests the equivalence hypothesis is not well enough developed to test such a hypothesis. In such cases, it may be more productive to focus on effect estimation and description rather than leaping to hypothesis tests ([Scheel et al. 2020](#)). Throughout the examples in this paper, equivalence bounds, and the data sets themselves, are chosen for illustrative purposes; readers should not treat them as direct recommendations for their own research.

### 3. TOST with t-tests

The `t.TOST` function provides a unified interface for equivalence and minimal effect testing with t-tests. It runs a standard null significance test ( $H_0: \mu = 0$ ) alongside the equivalence (or minimal effect) test, because in practice researchers may want both: “Is there a significant effect?” and “Can I rule out a meaningful effect?” The function handles independent, paired, and one-sample designs through the `paired` argument and accepts data via either a formula interface or raw vectors, making it a direct analog to base R’s `t.test`. Additionally, the output includes an SMD calculation alongside the unstandardized mean difference with appropriate, albeit approximate, confidence intervals. For researchers who only want to test a single hypothesis at a time, the `simple.htest` function (described in the next section) provides a lighter alternative.

Minimal effect testing (MET) is also supported. MET is useful when the hypothesis is that an effect *exceeds* a meaningful threshold — the null hypothesis is equivalence, and rejection implies a practically meaningful effect (see Figure 1) ([Mazzolari et al. 2022](#)).



**Figure 1.** Types of hypotheses tests supported by two one-sided tests (TOST) procedures.

### 3.1. *Independent Groups Example*

In these examples we will use the `sleep` data and the `bugs` data from the `jmv` R package.

```
data('sleep')
library(jmv)
data('bugs')
```

For the first example, we use the `sleep` data with equivalence bounds (set by the `eqb` argument) of  $\pm 0.5$  units. The `smd_ci` argument can also be set to use one of four possible calculations for the SMD confidence interval. In the example below, I have opted to go with the normal approximation for computational ease; such approximations tend to provide adequate coverage ([Viechtbauer 2007](#)). The formula interface works almost identically to `t.test`:

```
res1 = t_TOST(formula = extra ~ group, # difference in "group" of outcome "extra"
              data = sleep, # data
              eqb = .5, # equivalence bound
              smd_ci = "z") # sets type of SMD confidence interval calculation
```

The `print` method provides a verbose summary including the two-tailed test, both one-sided equivalence tests, and the standardized mean difference (Hedges'  $g$  by default):

```
print(res1)

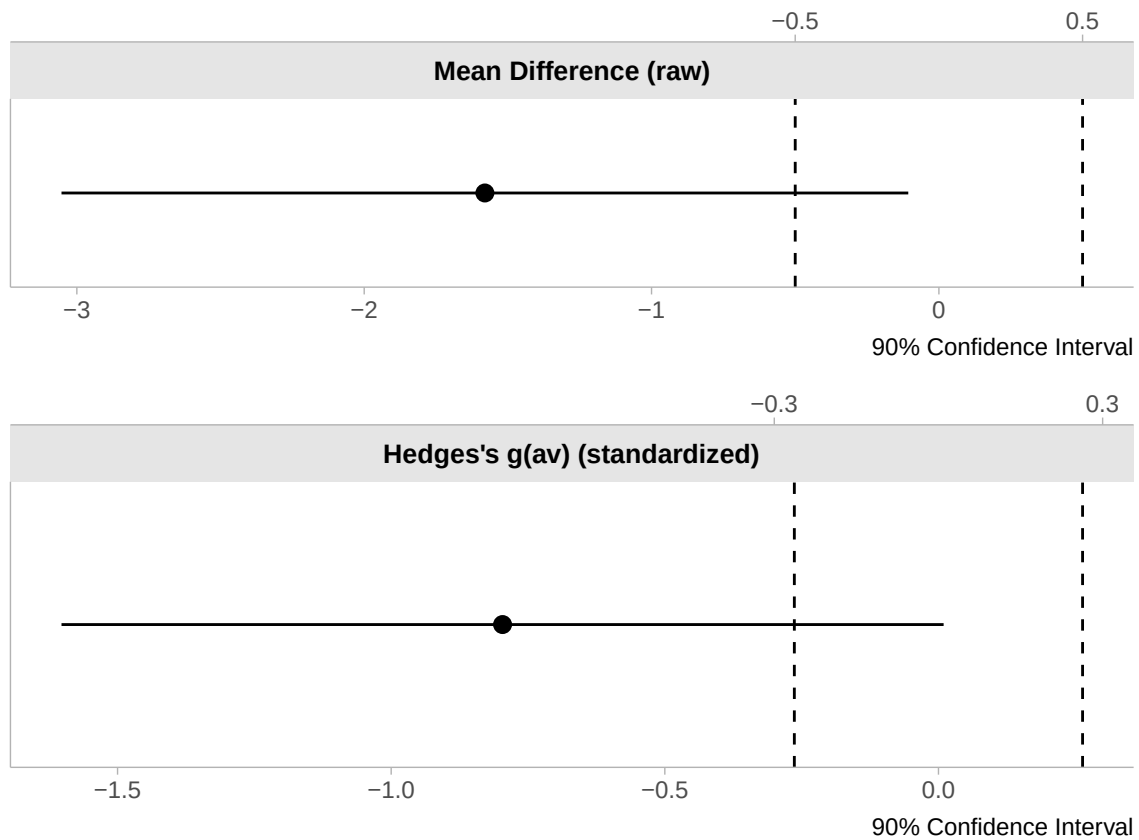
##
## Welch Two Sample t-test
##
## The equivalence test was non-significant, t(17.78) = -1.3, p = 0.89
```

```

## The null hypothesis test was non-significant,  $t(17.78) = -1.86$ ,  $p = 0.08$ 
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t      df p.value
## t-test    -1.861 17.78  0.079
## TOST Lower -1.272 17.78  0.890
## TOST Upper -2.450 17.78  0.012
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           -1.5800 0.8491 [-3.0534, -0.1066]      0.9
## Hedges's  $g(av)$  -0.7965 0.4900 [-1.6024, 0.0094]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").

```

The equivalence test was non-significant,  $t(17.78) = -1.3$ ,  $p = 0.89$   
The null hypothesis test was non-significant,  $t(17.78) = -1.86$ ,  $p = 0.08$   
Equivalence bounds:  $[-0.5, 0.5]$  (raw)



**Figure 2.** Results of the independent-samples equivalence test visualized with the default plot method. The top panel displays the raw mean difference with its 90% confidence interval, and the bottom panel displays the standardized effect size (Hedges's  $g_{av}$ ) with its 90% confidence interval. In both panels, the dashed vertical lines indicate the equivalence bounds (raw: -0.5 to 0.5; standardized: approximately -0.3 to 0.3), and the point estimate is marked by a filled circle. A 90% confidence interval is used because it corresponds to the two one-sided tests (TOST) procedure at an alpha level of 0.05.

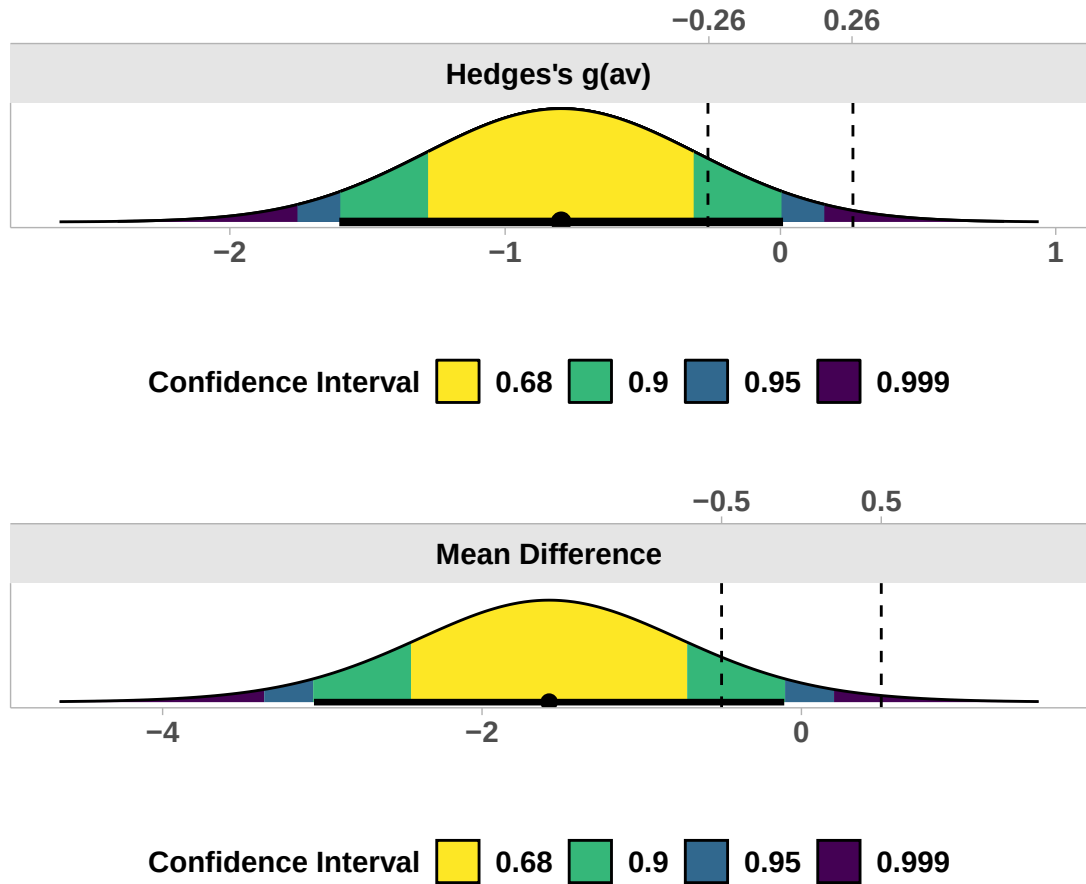
The default visualization provided by the generic `plot` method for `t_TOST` provides a simple visualization of both effect sizes and their respective confidence intervals alongside the equivalence bounds. Test information is also printed at the top of the plot.

```
plot(res1)
```

Also when the generic `plot` method has the `type` argument set to "cd" this produces a consonance density plot, inspired by the "concurve" R package (Rafi and Greenland 2020), that shows the estimate, confidence intervals, and equivalence bounds simultaneously:

```
plot(res1, type = "cd")
```

Other plot types are available: `type = "c"` produces a consonance plot showing



**Figure 3.** Consonance density plot for the independent-samples equivalence test. The top panel displays the standardized effect size (Hedges's  $g_{av}$ ) and the bottom panel displays the raw mean difference. In each panel, the consonance density curve represents the distribution of confidence intervals across all levels, with color-coded regions corresponding to the 68%, 90%, 95%, and 99.9% confidence intervals. The point estimate is marked by a filled circle along the x-axis. The dashed vertical lines indicate the equivalence bounds (standardized: -0.26 to 0.26; raw: -0.5 to 0.5). This visualization conveys the full range of effect sizes compatible with the data at varying confidence levels, rather than relying on a single confidence interval. Wider, less certain intervals (e.g., 99.9%) extend further from the point estimate, while narrower intervals (e.g., 68%) cluster near it.

multiple confidence interval levels, and shading can be customized with the `ci_shades` argument.



### 3.2. One-Sample t-test with Asymmetric Bounds

When the equivalence bounds are not symmetric around the null, two values can be supplied to `eqb`. This is common when the research question implies asymmetric consequences. Similarly, we may want to test the mean itself rather than a mean difference. This harkens back to the original use of the t-test by Gosset at Guinness Brewery, where the concern was whether a single sample mean fell within acceptable limits. Here we test whether the mean of LDHF falls within the bounds of 5.5 and 8.5 around a hypothesized value of 7.5:

```
res4 = t.TOST(x = bugs$LDHF, # simple sample, vector from which to estimate mean
              mu = 7.5, # set nil significance test to 7.5
              eqb = c(5.5,8.5) # set TOST bounds to 5.5 and 8.5
              )
res4

##
## One Sample t-test
##
## The equivalence test was significant, t(90) = -4.2, p < 0.01
## The null hypothesis test was non-significant, t(90) = -0.458, p = 0.65
## NHST: don't reject null significance hypothesis that the effect is equal to 7.5
## TOST: reject null equivalence hypothesis
##
## TOST Results
##               t df p.value
## t-test       -0.4577 90  0.648
## TOST Lower    7.1156 90 < 0.001
## TOST Upper   -4.2444 90 < 0.001
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw         -0.12088 0.2641  [6.9402, 7.818]      0.9
## Hedges's g  -0.04758 0.1049 [-0.2185, 0.1236]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

The same interface handles paired designs by setting `paired = TRUE`. Standardized mean differences are reported automatically; Hedges's correction is applied by default but can be toggled with `bias_correction`, and Glass's delta can be requested via the `glass` argument. Setting equivalence bounds in SMD units (via `eqbound_type = "SMD"`) is possible but discouraged because the bound becomes dependent on sample variance (see the Standardized Effect Sizes section below).

### 3.3. Checking Assumptions

The parametric t-tests implemented in `t.TOST` assume independent observations, approximately normal residuals (or differences, in the paired and one-sample cases), and, for Student's form, equal variances between groups; Welch's correction relaxes the equal-variance assumption and is the default. The residual normality assumption matters most in small samples, where the central limit theorem provides limited protection. Visual diagnostics are generally more informative than formal tests: QQ plots

of residuals or differences for normality, side-by-side boxplots or density overlays for variance comparison, and inspection of residual plots or boxplots for extreme or influential observations. A worked diagnostic workflow for each design is provided in the online supplement. When assumptions are seriously in doubt, the bootstrap and permutation alternatives discussed in later sections provide valid inference under relaxed — though not eliminated — distributional requirements.

### 3.4. Using Summary Statistics

When only summary statistics are available (e.g., from a published article), the `tsum.TOST` function performs the same tests using sample means, standard deviations, sample sizes, and (for paired designs) the correlation between measures<sup>2</sup>:

```
res_tsum = tsum_TOST(
  m1 = mean(bugs$LDHF, na.rm=TRUE),
  sd1 = sd(bugs$LDHF, na.rm=TRUE),
  n1 = length(na.omit(bugs$LDHF)),
  hypothesis = "EQU", smd_ci = "t",
  eqb = c(5.5, 8.5)
)

res_tsum

##
## One-sample t-test
##
## The equivalence test was significant, t(90) = -4.244, p = 2.66e-05
## The null hypothesis test was significant, t(90) = 27.942, p = 3.91e-46
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##
##           t df p.value
## t-test      27.942 90 < 0.001
## TOST Lower   7.116 90 < 0.001
## TOST Upper  -4.244 90 < 0.001
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           7.379 0.2641 [6.9402, 7.818]         0.9
## Hedges's g     2.905 0.2395 [2.4289, 3.3804]         0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

## 4. Simplified Hypothesis Testing

The `t.TOST` function is designed to be comprehensive in that it runs three tests, computes standardized effect sizes, and produces verbose output. This is useful for exploration or situations where researchers *always* want to pair a traditional nil significance test with an equivalence test. However, sometimes a researcher has a single specific hypothesis to test, or simply wants less verbose output (e.g., SMD calculation is not wanted). The `simple_hstest` function provides a lighter interface for this purpose. It returns a standard R `hstest` object — the same class returned by `t.test()` and `wilcox.test()` — but with expanded `alternative` options that include "equivalence" and "minimal.effect". This means researchers can perform

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<sup>2</sup>The `extract_r_paired` function can be used if the correlation between paired observations is not readily available.

equivalence testing with the same familiar interface they already use for standard hypothesis tests<sup>3</sup>.

Alongside `simple_htest`, a set of helper functions makes it easy to work with any `htest` object: `as_htest` converts verbose TOST output to the simpler format, `describe_htest` generates formatted text for reporting, `df_htest` converts results to data frames for tabulation, and `plot_htest_est` produces estimate plots with equivalence bounds. Checking assumptions would follow the same workflow as `t.TOST`, as both are using t-tests, as discussed in the online supplement.

#### 4.1. *The simplified function*

The `simple_htest` function supports `t.test` and `wilcox.test` as the underlying test (via the `test` argument) and all five alternative hypotheses: `"two.sided"`, `"less"`, `"greater"`, `"equivalence"`, and `"minimal.effect"`. The `mu` parameter specifies equivalence bounds — a single value creates symmetric bounds (e.g., if 1 is provided at `mu` with `alternative = "equivalence"` then the equivalence region will be  $[-1,1]$ ), or two values can be given for asymmetric bounds.

```
res_sh1 <- simple_htest(extra ~ group,
                        data = sleep,
                        paired = TRUE,
                        alternative = "equivalence",
                        mu = 0.5)
```

```
res_sh1
```

```
##
## Paired t-test
##
## data: extra by group
## t = -2.7766, df = 9, p-value = 0.9892
## alternative hypothesis: equivalence
## null values:
## mean difference mean difference
##          -0.5          0.5
## 90 percent confidence interval:
## -2.2930053 -0.8669947
## sample estimates:
## mean of the differences (z = '1' - '2')
##                                -1.58
```

Additionally, tests other than TOST can be applied with this function. The traditional nil-hypothesis tests, non-inferiority, or even superiority-by-a-margin hypotheses can all be tested with this function. For example, let's say we want to test that a treatment group in the sleep data was just non-inferior to the other treatment rather than equivalent. This can be accomplished by just setting the direction with the `alternative` argument and the margin of non-inferiority by the `mu` argument.

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<sup>3</sup>In fact, users can mostly replace `t.test()` and `wilcox.test()` with `simple_htest` as it also allows for the standard alternative hypotheses of two-sided, greater, less. If you have a hypothesis that would traditionally utilize a t-test or Wilcoxon-Mann-Whitney (WMW) test, then you can jump right to using `simple_htest` in TOSTER.

```

res_sh2 <- simple_hstest(extra ~ group,
                        data = sleep,
                        paired = TRUE,
                        alternative = "greater",
                        mu = -0.5)

res_sh2

##
## Paired t-test
##
## data: extra by group
## t = -2.7766, df = 9, p-value = 0.9892
## alternative hypothesis: true mean difference is greater than -0.5
## 95 percent confidence interval:
## -2.293005 Inf
## sample estimates:
## mean of the differences (z = '1' - '2')
## -1.58

```

Notice how the output, in either hypothesis test above, is a single clean `hstest` object. This is much more concise than the verbose `t.TOST` output shown previously on purpose. If you want to perform additional tests — such as a complementary nil significance tests — then that has to be done separately. However, there are some methods (described below) that make this `hstest` format easy to deal with when reporting statistical test results.

#### 4.2. Describing and Tabulating Results

For writing up results, `describe_hstest` generates a formatted text string following typical scientific reporting conventions:

```

describe_hstest(res_sh1)

## [1] "The Paired t-test is not statistically significant (t(9) = -2.78, p = 0.989, mea

```

#### 4.3. Plotting Results

A visual summary of the estimate and confidence interval can be produced with `plot_hstest_est`. For equivalence and minimal effect tests, the equivalence bounds are shown as dashed vertical lines:

```

plot_hstest_est(res_sh1)

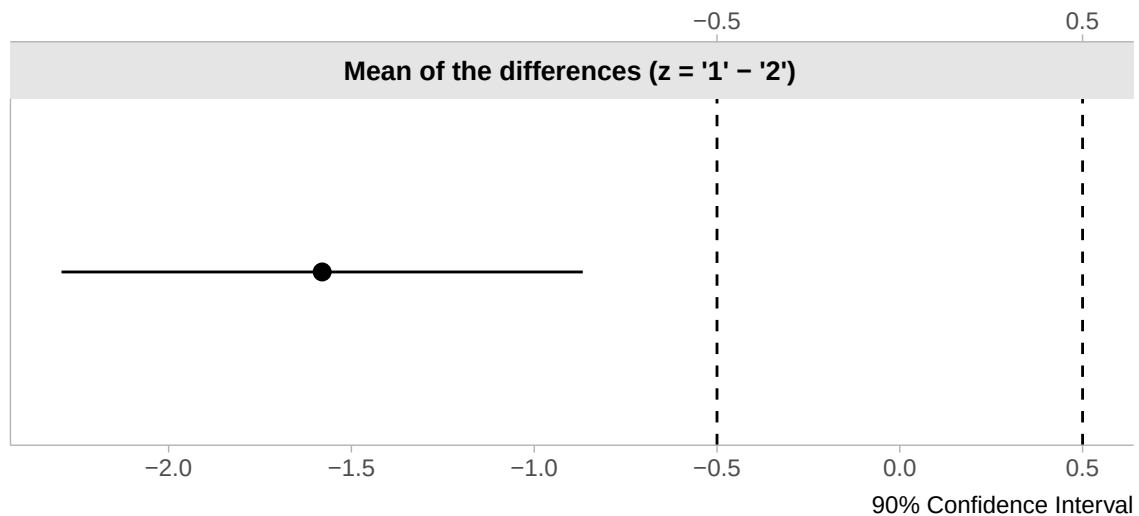
```

### Paired t-test

$t(9) = -2.78, p = 0.989$

mean of the differences ( $z = '1' - '2'$ ) =  $-1.58$ , 90% CI  $[-2.29, -0.867]$

null: mean difference  $< -0.5$  or  $> 0.5$



**Figure 4.** Estimate plot for an equivalence test via `simple_htest`.

## 5. Standardized Effect Sizes

This section provides a unified treatment of the standardized effect size tools in TOSTER — standardized mean differences (SMDs), correlation coefficients, and tools for comparing both between studies. Before demonstrating these tools, it is worth discussing the limitations of standardized effect sizes, because these limitations are especially important for equivalence testing.

### 5.1. Problems with Standardized Effect Sizes

Standardized effect sizes — Cohen’s  $d$ , correlation coefficients, standardized regression coefficients — are appealing because they allow comparison across studies and scales, are required by many journals, and serve as the primary currency of meta-analysis. However, they have well-documented problems that are especially relevant to equivalence testing.

Standardized mean differences are sensitive to the sample variance. The “same” real-world effect produces different  $d$  values across populations and study designs. The ubiquitously reported Cohen’s  $d$  conflates the magnitude of the mean difference with the precision of measurement, which means that a narrow equivalence bound on  $d$  may correspond to a very different raw-scale bound depending on the variability of the sample (Caldwell and Vigotsky 2020; Baguley 2009). This makes it unclear what, substantively, is being declared “equivalent” — are we claiming the means are close, or that the means are close *relative to the variance*? The choice of denominator (pooled SD, control SD, within-subject SD) further complicates that same dubious interpretation.

Correlations have analogous issues. Tukey (1969) noted that correlations are in-

fluenced by the range of both variables, so the “same” bivariate relationship produces different coefficients depending on sample selection. A correlation of 0.3 in a restricted-range sample may correspond to a much larger association in the full population. Standardized regression coefficients share these vulnerabilities. These depend on the variances of both the predictor and outcome, which makes them unstable across samples and difficult to interpret substantively (Greenland et al. 1991).

The implication for equivalence testing is straightforward. When the measurement scale is meaningful, researchers should generally prefer *unstandardized* (raw-scale) equivalence bounds. When standardized bounds are used, the choice should be justified with reference to the specific population and measurement context, not borrowed from generic benchmarks.

With these caveats in mind, the following subsections describe the tools TOSTER provides for computing and testing standardized effects assuming they are appropriate.

## 5.2. Standardized Mean Difference Functions

The `smd_calc` function computes standardized mean differences — Cohen’s  $d$ , Hedges’  $g$  (bias-corrected), or Glass’s  $\Delta$  — with confidence intervals and *optional* hypothesis testing (utilizing a z-test:  $z = (d - \delta_0)/SE_d$ , where  $d$  is the sample SMD estimate,  $\delta_0$  is the null value being tested, and  $SE_d$  is its standard error.). It supports one-sample, paired, and two-sample designs, and can return either an `htest` object or a data frame. The SMD is already reported in `t.TOST` output, but `smd_calc` allows standalone computation and supports equivalence testing directly on the effect size. This may be useful as a complementary function when using `simple_htest` as the primary statistical testing function.

When estimation rather than testing is the goal, `smd_calc` can be called with `alternative = "none"` (the default), which returns only the point estimate and confidence interval without performing a hypothesis test. This is the recommended usage when the primary objective is to report the magnitude of the effect.

```
smd_calc(formula = extra ~ group,
          data = sleep,
          paired = TRUE,
          bias_correction = TRUE,
          rm_correction = TRUE,
          output = "htest")
```

```
##
## Paired Sample Standardized Mean Difference (SMD; Hedges's
## g[rm]=(‘1’-‘2’)/SD_rm)
##
## data:  extra by group
##
## alternative hypothesis: none
## 95 percent confidence interval:
## -1.39802439 -0.07479266
## sample estimates:
## SMD (g[rm])
## -0.7513666
```

When hypothesis testing on the SMD is desired, `smd_calc` supports equivalence and minimal-effect alternatives directly:

```
smd_calc(formula = extra ~ group,
          data = sleep,
          paired = TRUE,
          bias_correction = TRUE,
          rm_correction = TRUE,
          alternative = "equivalence",
          null.value = c(-0.5, 0.5),
          output = "htest")

##
## Paired Sample Standardized Mean Difference (SMD; Hedges's
##  $g[rm] = (d_1 - d_2) / SD_{rm}$ 
##
## data: extra by group
## z = -0.78495, p-value = 0.7838
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
## -0.5 0.5
## 90 percent confidence interval:
## -1.2894564 -0.1789204
## sample estimates:
## SMD ( $g[rm]$ )
## -0.7513666
```

However, the parametric z-test used by `smd_calc` relies on distributional assumptions about the sampling distribution of the SMD that may not hold in practice. When a hypothesis test on the SMD is the goal, bootstrapping via `boot_smd_calc` is recommended because it provides more robust confidence intervals and p-values.

```
boot_smd_calc(formula = extra ~ group,
               data = sleep,
               paired = TRUE,
               bias_correction = TRUE,
               rm_correction = TRUE,
               alternative = "equivalence",
               null.value = c(-0.5, 0.5))

##
## Paired Sample bootstrapped Standardized Mean Difference (SMD; Hedges's
##  $g[rm] = (d_1 - d_2) / SD_{rm}$ 
##
## data: extra by group
## z-observed = -0.78495, p-value = 0.8934
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
```



```
##          -0.5          0.5
## 90 percent confidence interval:
## -1.2978837 -0.4361528
## sample estimates:
## SMD (g[rm])
## -0.7513666
```

### 5.2.1. Checking Assumptions

The SMD functions inherit the assumptions of the underlying t-test: independent observations, approximately normal residuals or differences, and, for two-sample designs, comparable variances across groups. Visual diagnostics from the t-test discussion above apply directly. Two considerations are specific to standardized effect sizes, however. First, outliers and skew affect both the numerator (mean difference) and the denominator (standard deviation) of the SMD, which can amplify their influence on the estimate; the `tr` trimmed-means argument available in the `boot_smd_calc` function is particularly worth considering when the raw data show extreme values. Second, the asymptotic z-test in `smd_calc` relies on an approximation to the sampling distribution of the SMD that can be very inaccurate in small samples, which is the primary motivation for preferring `boot_smd_calc` when hypothesis testing is the goal.

### 5.3. Equivalence Tests for Correlations

Researchers sometimes need to test that a correlation is practically zero. The `z_cor_test` function provides a asymptotic approach using the Fisher z-transformation. However, this is only an approximate method and is generally *not recommended* as the primary inferential tool for equivalence testing on correlations.

```
z_cor_test(
  x = bugs$LDLF,
  y = bugs$LDHF,
  method = "pearson",
  alternative = "t")

##
## Pearson's product-moment correlation with approximate SE
##
## data: bugs$LDLF and bugs$LDHF
## z = NA, N = 91, p-value = NA
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## NA NA
## sample estimates:
## r
## NA
```

Because the z-test relies on distributional approximations that can be inaccurate — particularly with small samples — bootstrapping via `boot_cor_test` is recommended for more robust inference. The `boot_cor_test` function supports all three major correlation methods (Pearson, Kendall, and Spearman) as well as the Winsorized ([Wilcoxon](#)

1993) and percentage bend (Wilcox 1994) correlation coefficients as robust alternatives, and returns standard `htest` objects that integrate with the helper functions described above.

```
res_boot_cor = boot_cor_test(  
  x = bugs$LDLF,  
  y = bugs$LDHF,  
  method = "pearson",  
  alternative = "equivalence",  
  null = 0.4)  
  
print(res_boot_cor)
```

```
##  
## Bootstrapped Pearson's product-moment correlation (BCa)  
##  
## data: bugs$LDLF and bugs$LDHF  
## N = 91, p-value = 0.9825  
## alternative hypothesis: equivalence  
## null values:  
## correlation correlation  
## 0.4 -0.4  
## 90 percent confidence interval:  
## 0.4419166 0.6857747  
## sample estimates:  
## r  
## 0.5767783
```

```
describe_htest(res_boot_cor)
```

```
## [1] "The Bootstrapped Pearson's product-moment correlation (BCa) is not statistically
```

If only summary statistics are available, an approximate method is provided through `corsum_test`, which uses the Fisher z-transformation to compute confidence intervals and p-values.

### 5.3.1. Checking Assumptions

Correlation-based inference in `z_cor_test` and `boot_cor_test` assumes independent paired observations and, for Pearson's coefficient, a roughly linear relationship between variables with approximately symmetric marginal distributions; Spearman and Kendall relax the linearity requirement to monotonicity, making them appropriate when the relationship is non-linear but directionally consistent. The bootstrap version in `boot_cor_test` further relaxes distributional assumptions about the marginals but remains sensitive to observations that exert disproportionate influence on the estimate. Visual diagnostics are the primary tools: scatterplots with a loess smoother to assess linearity or monotonicity, marginal density plots or QQ plots to check distributional shape, and jackknife-after-bootstrap plots to identify individual observations driving bootstrap variability. A worked diagnostic workflow is provided in the online supplement.

#### 5.4. Comparing Standardized Effect Sizes Between Studies

When evaluating whether a replication study produced results consistent with the original, researchers need a way to formally compare effect sizes. The `compare_smd` function tests whether two SMD estimates — from an original study and a replication, for example — differ more than would be expected by sampling variability alone.

Consider an original study reporting Cohen's  $d_z = 0.95$  with 25 participants and a replication with  $d = 0.23$  and 50 participants. Are these estimates compatible?

```
compare_smd(smd1 = 0.95,
            n1 = 25,
            smd2 = 0.23,
            n2 = 50,
            paired = TRUE,
            null = .25,
            alternative = "equivalence")

##
## Difference in Cohen's dz (paired)
##
## data: Summary Statistics
## z = 1.6767, p-value = 0.9532
## alternative hypothesis: equivalence
## null values:
## difference in SMDs difference in SMDs
##                0.25                -0.25
## sample estimates:
## difference in SMDs
##                0.72
```

Based on these results, we would reject the null significance hypothesis (the SMDs differ significantly) but fail to reject the equivalence null hypothesis (we cannot conclude the SMDs are practically equivalent). When raw data are available, `boot_compare_smd` provides a bootstrap alternative (see the package vignettes for details).

For comparing correlation coefficients between independent studies, the `compare_cor` function provides an analogous test. It uses either the Fisher  $z$ -transformation (default) or the Kraatz (Anderson-Hauck) method to compare two correlation coefficients.

```
compare_cor(r1 = 0.45,
            df1 = 48,
            r2 = 0.25,
            df2 = 78,
            alternative = "equivalence",
            null = 0.25)

##
## Difference between two independent correlations (Fisher's z transform)
##
## data: Summary Statistics
```

```
## z = -0.026125, p-value = 0.4439
## alternative hypothesis: equivalence
## null values:
## difference between correlations difference between correlations
##                                0.25                                -0.25
## sample estimates:
## difference between correlations
##                                0.2
```

When raw data are available for both studies, `boot_compare_cor` provides a bootstrap alternative that avoids the distributional approximations underlying the Fisher z and Kraatz methods.

The comparison functions `compare_smd` and `compare_cor` inherit the assumptions of the underlying effect size estimates they combine. Because both studies contribute to the comparison, violations in either study propagate into the test. When raw data are available, the bootstrap alternatives (`boot_compare_smd` and `boot_compare_cor`) are preferred because they avoid the Fisher z and related asymptotic approximations, which can be inaccurate particularly when effect sizes are large or sample sizes are modest. When only summary statistics are available, readers should report the diagnostic context of the original and replication studies where possible.

**A note on argument names for equivalence bounds.** The equivalence bound argument is named differently across TOSTER functions for historical and compatibility reasons: `eqb` in `t.TOST`, `wilcox.TOST`, `equ.ftest`, and `equ.anova`; `mu` in `simple.htest`, `brunnermunzel`, `boot.t.test`, and `perm.t.test` (for compatibility with base R's `t.test`); and `null` in `compare.smd` and `z.cor.test`. Despite the different names, the interpretation is the same: the bound(s) defining the region of practical equivalence.

## 6. Robust Methods for Equivalence Testing

When t-test assumptions (normality, homoscedasticity) are seriously violated, or when the mean difference is a poor summary of the effect (e.g., with heavy skew or ordinal psychometric scales) alternative approaches are worth considering. TOSTER organizes these into three categories: rank-based tests, resampling methods, and log-transformed methods.

### 6.1. Rank-Based Tests

Rank-based tests operate on the ranks of the data rather than the raw values, which is why they are typically described as non-parametric. It is worth emphasizing that this label applies to the test, not the data. The underlying observations may still come from any distribution, and rank-based methods should not be treated as automatic substitutes for a t-test when normality is in doubt. These tests are not tests of means or medians (Divine et al. 2018; Karch 2021); they target different estimands (i.e., the quantity a study is attempting to estimate), and the choice between a rank-based test and a t-test should be driven by which estimand is of scientific interest.

#### 6.1.1. Wilcoxon-Mann-Whitney Tests

The WMW group of tests provide non-parametric tests based on ranks. Within the TOST framework, two directional location-shift tests determine if the shift is within (equivalence) or outside (minimal effect) the equivalence bounds.

However, there are important caveats. Many researchers mistakenly believe these are tests of medians, but without a location-shift assumption, the WMW test is a test of stochastic equality. As Fay and Malinovsky (2018) demonstrate, the effect size can even be presented as stochastic superiority/dominance. Only under a location-shift assumption does it become a test of central tendency through the Hodges-Lehmann estimator. Divine et al. (2018) demonstrate striking counterexamples: groups can have equal medians yet produce a significant WMW test, very different medians yet a non-significant WMW test, or even significance in the opposite direction from the median comparison.

For equivalence testing, bounds are applied to the pseudo-median, which corresponds to the mean or median difference only under the location-shift assumption. For testing hypotheses about mean differences, `perm.t.test` or `boot.t.test` are more appropriate. For stochastic superiority, `brunnermunzel` provides a clearer framework.

```
test1 = wilcox.TOST(formula = extra ~ group,
                    data = sleep,
                    paired = TRUE,
```

```

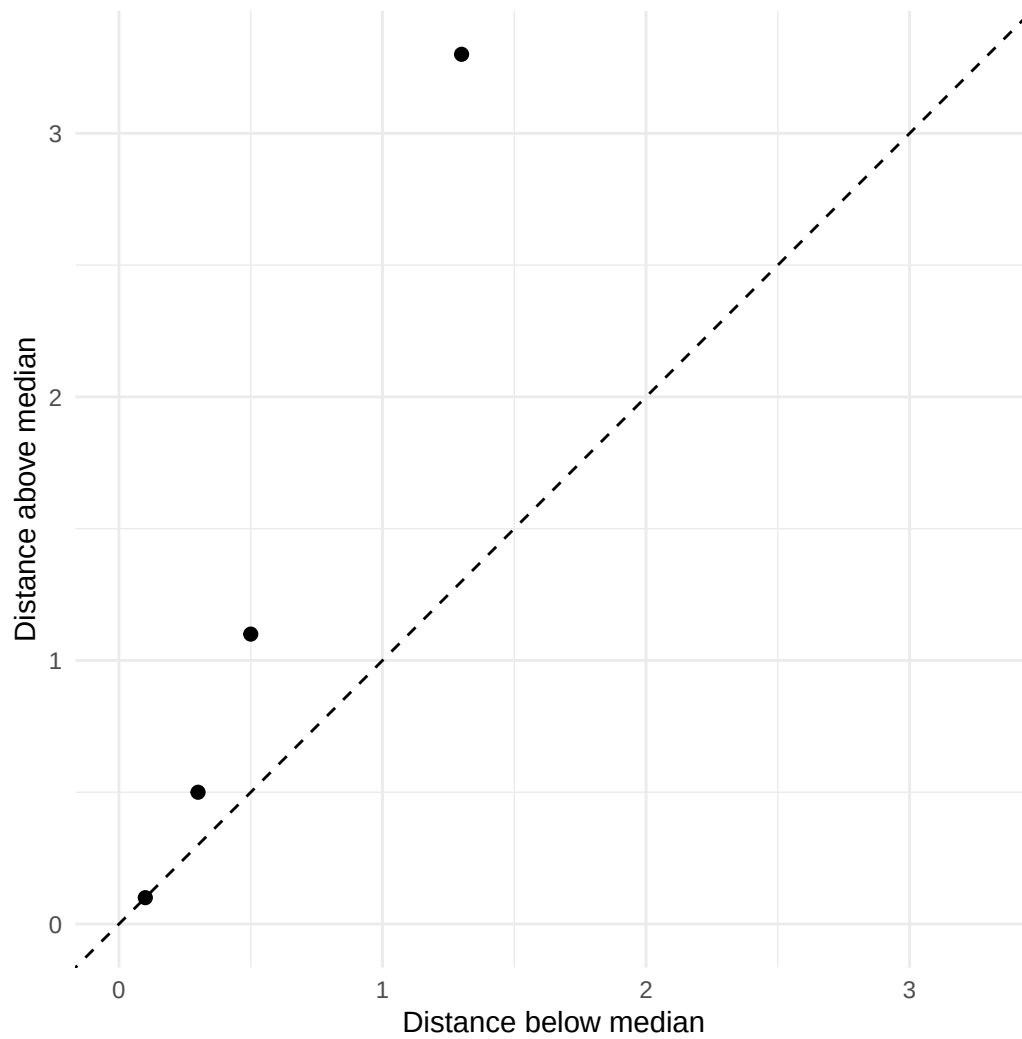
                                eqb = .5)
print(test1)

##
## Wilcoxon signed rank test with continuity correction
##
## The equivalence test was non-significant V = 0.000, p = 9.96e-01
## The null hypothesis test was significant V = 0.000, p = 9.09e-03
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           Test Statistic p.value
## NHST                0    0.009
## TOST Lower           2    0.996
## TOST Upper           0    0.003
##
## Effect Sizes
##           Estimate      C.I. Conf. Level
## Median of Differences    -1.4 [-2.7, -1.15]      0.9
## Rank-Biserial Correlation -1.0 [-1, -0.4492]      0.9

```

Based on these results, we would conclude there is a significant difference and the results are also not equivalent. But again, the estimate we are testing is the pseudo-median, not the mean or median difference, so this conclusion may not be relevant to the research question at hand.

**6.1.1.1. Checking Assumptions.** The WMW tests assume independent observations and exchangeability under the null, the latter meaning that group labels carry no information about outcomes if there is no effect. For the interpretations discussed above to be clean, two additional assumptions matter. The location-shift assumption — that the two group distributions differ only in location, not shape — is what allows WMW results to be interpreted as a test of the median difference via the Hodges-Lehmann estimator. Violations are best diagnosed by overlaying the empirical cumulative density functions of the two groups; under location shift, the curves are horizontal translations of each other, whereas crossing curves or varying horizontal gaps indicate violation. For the paired and one-sample signed-rank variants used in `wilcox.TOST` or `simple.htest`, the signed-rank procedure additionally requires that the difference distribution is symmetric around the null value for the pseudo-median to coincide with the mean or median difference. Asymmetry is readily checked with a symmetry plot of the paired differences (Figure 5). As with the location-shift case, visible departures suggest that the test's output — the pseudo-median and associated p-value — does not map cleanly onto a hypothesis about central tendency.



**Figure 5.** Symmetry plot of the paired differences from the sleep dataset, pairing each observation's distance above the median with the corresponding distance below. Under symmetry, points fall along the 45-degree reference line (dashed); systematic departures — points consistently above or below the line — indicate asymmetry. The signed-rank variant of WMW requires this symmetry for the pseudo-median to coincide with the mean or median of the differences.

### 6.1.2. Brunner-Munzel Test

The Brunner-Munzel test (Brunner and Munzel 2000) estimates the relative effect  $\hat{p} = P(X > Y) + 0.5 \cdot P(X = Y)$ , sometimes referred to as the probability of superiority — the probability that a randomly chosen observation from one group exceeds a randomly chosen observation from the other, with ties split evenly. This estimand is interpretable without distributional assumptions, unlike the WMW test which requires a location-shift assumption to say anything about location differences. Karch (2021) argues it should be preferred as the default non-parametric procedure because it is valid under heteroscedasticity and provides an interpretable effect size.

Stochastic superiority tests may be particularly well-suited to Likert-type scales which are ubiquitous in psychological research. For a single ordinal item, the spacing between response categories carries no guaranteed meaning, and a mean difference therefore has a strained interpretation. For example, what, concretely, does a 0.4-unit mean difference on a five-point scale communicate? The stochastic superiority estimand sidesteps this entirely. The probability that a randomly selected participant from one group outscores a randomly selected participant from the other is interpretable on its own terms, with no knowledge of the scale required. We are not suggesting that t-tests will *always* mislead in these contexts but it is a concern that should not be quickly dismissed either (Liddell and Kruschke 2018). Additionally, even if we assume traditional parametric methods are robust to the ordinal nature of the data, the preserved error rate is not the same as interpretive clarity. For researchers whose primary outcomes live on ordinal scales, the Brunner-Munzel test and its associated effect size — as well as ordinal models more generally (Bürkner and Vuorre 2019) — may offer a more natural and defensible default.

The `brunner_munzel` function supports equivalence and minimal effect testing via `alternative`, with bounds on the probability scale via `mu`. Results can be reported on different scales ("probability", "difference", "logodds", or "odds"), and three test methods are available: "t" (default), "logit" (range-preserving CIs), and "perm" (studentized permutation; Neubert and Brunner (2007)).

```
bm_test = brunner_munzel(formula = extra ~ group,
                          data = sleep,
                          alternative = "equivalence",
                          mu = c(0.3, 0.7))

print(bm_test)
```

```
##
## Two-sample Brunner-Munzel test
##
## data: extra by group
## t = -0.39392, df = 16.898, p-value = 0.6507
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
##          0.3          0.7
## 90 percent confidence interval:
## 0.0562039 0.4537961
## sample estimates:
## P('1'>'2') + .5*P('1'='2')
```



**6.1.2.1. Checking Assumptions.** The Brunner-Munzel test has unusually light assumption requirements. It requires independent observations within and between groups, and (for the t-approximation used by default) sample sizes sufficient for the asymptotic approximation to be reasonable. That's essentially it. Unlike WMW, there is no location-shift requirement for the test to be interpretable as a hypothesis about location, and unlike the proportional-odds framing of WMW as a score test from an ordinal regression model, no constant-odds-ratio assumption is needed. The relative effect is defined directly from the two marginal distributions and carries the same interpretation — the probability that a random observation from one group exceeds one from the other — regardless of whether the distributions are symmetric, skewed, unimodal, or shaped however they happen to be. Visual inspection of the group distributions remains useful for understanding the data and for choosing a reporting scale, but there is no distributional assumption that these checks are verifying. The studentized permutation variant (`method = "perm"`) is worth considering in small samples, where the t-approximation may be less trustworthy.

## 6.2. Resampling Methods

When the goal is to test hypotheses about means while relaxing distributional assumptions, resampling methods provide powerful tools. TOSTER implements both bootstrap and permutation approaches through `boot.t.test` and `perm.t.test`, each with distinct inferential logic but a shared motivation to retain the mean as the estimand while obtaining reference distributions empirically — through resampling — rather than relying on the normal-theory derivations (i.e., Student’s t-test).

Both approaches are natural extensions of the classic t-test rather than alternatives to it. The test statistic itself is unchanged; what changes is how the null distribution is constructed. In the conventional t-test, the null distribution of the standardized mean difference is derived analytically under the assumption that the data are drawn from a normal distribution, with equal variances assumed in Student’s t-test and unequal variances accommodated in Welch’s t-test. The bootstrap approximates the sampling distribution of the same statistic by resampling from the observed data with replacement, whereas the permutation approach constructs an exact or near-exact null distribution by shuffling group labels (two-sample) or flipping signs (one-sample and paired). In both cases, the researcher continues to ask the same question about means; only the calibration of the reference distribution differs. This is important for researchers who may view resampling methods as a departure from familiar testing logic: the estimand, the null hypothesis, and the test statistic remain the same.

Because the reference distribution is generated from the data rather than assumed, distributional requirements are relaxed, though not eliminated. The bootstrap relies on asymptotic arguments and assumes that the empirical distribution is a reasonable stand-in for the population distribution; it performs well with moderate-to-large samples and skewed or heavy-tailed data, but can struggle with very small samples or extreme outliers. The permutation approach is exact under the null hypothesis of exchangeability, meaning group labels carry no information about outcomes. In its basic form this exactness requires identical distributions under the null, which implicitly assumes equal variances. The studentized permutation test ([Janssen 1997](#); [Chung and Romano 2013](#)) addresses this limitation by permuting a scale-invariant statistic (essentially the Welch t-statistic), yielding asymptotically valid inference even under heteroscedasticity. [Arboretti, Pesarin, and Salmaso \(2020\)](#) extended this logic to the equivalence testing setting, showing that studentized permutation TOST maintains nominal Type I error rates when variances differ across groups. In short, these methods are assumption-lighter rather than assumption-free, and users should still consider whether independence and exchangeability are reasonable for their design.

A common reflex when normality seems implausible is to switch to the WMW or other rank-based tests. However, these tests answer a different question. As discussed earlier in this manuscript, rank-based procedures test hypotheses about stochastic ordering or pseudo-medians rather than means, and under heteroscedasticity they can reject the null even when the groups have identical means. When the scientific question is genuinely about means — as it often is in applied research where effect sizes and interventions are naturally expressed on the original scale — bootstrap and permutation t-tests should generally be preferred over rank-based methods for handling such violations of assumptions. The rank-based tests remain useful, but as answers to different questions, not as robust substitutes for tests of means.

Choosing between the bootstrap and permutation approaches involves tradeoffs rather than a clear winner in most situations. The studentized permutation test has the strongest theoretical footing for two-sample mean comparisons under heteroscedas-

ticity, and is particularly attractive for equivalence testing given the [Arboretti, Pesarin, and Salmaso \(2020\)](#) results. The bootstrap offers greater flexibility because it extends naturally to statistics beyond the mean difference, provides readily interpretable confidence intervals via CI inversion ([Thulin 2024](#)), and ensures consistency between reported intervals and p-values. In small samples, permutation approaches tend to have better finite-sample calibration because the null distribution is constructed exactly (or nearly so) from the data at hand, whereas the bootstrap's asymptotic justification can produce liberal tests with very small sample sizes. When outliers are a concern, both methods accept the `tr` argument for trimmed means, which reduces the influence of extreme observations while preserving the mean-based interpretation. As a rough heuristic, researchers comparing two independent groups with suspected unequal variances may find the studentized permutation test most defensible, while those wanting a single framework that yields coherent intervals and p-values across a range of designs may prefer the bootstrap. Both are substantial improvements over ignoring assumption violations, and both return standard `htest` objects that integrate with the TOST framework for equivalence and minimal effect testing.

### 6.2.1. Bootstrap t-test

The bootstrap t-test in TOSTER defaults to the studentized bootstrap (`boot.ci = "stud"`) because this matches the original bootstrap t-test proposed by [Efron and Tibshirani \(1993\)](#) (Chapter 16), which is the procedure `boot.t.TOST` implements. In this approach, the t-statistic itself is resampled rather than the raw mean difference, and confidence intervals are constructed from the bootstrap distribution of these pivotal quantities. Other CI methods (percentile, basic, and BCa) remain available via the `boot.ci` argument, but the studentized default preserves consistency with the canonical Efron and Tibshirani formulation. It also ensures that the reported p-value and confidence interval are derived from the same bootstrap distribution, so the two will always agree; previous versions of the package used the studentized p-value regardless of the selected CI method, which could produce disagreement between reported intervals and p-values.

```
set.seed(4522)
boot.t_test(formula = extra ~ group,
             data = sleep,
             paired = TRUE,
             alternative = "equivalence",
             mu = c(-0.5, 0.5),
             R = 999)

##
##  Bootstrapped Paired t-test (studentized)
##
## data:  extra by group
## t-observed = -2.7766, df = 9, p-value = 1
## alternative hypothesis: equivalence
## null values:
## mean difference mean difference
##           -0.5           0.5
## 90 percent confidence interval:
```

```
## -2.853166 -1.071659
## sample estimates:
## mean of the differences (z = '1' - '2')
## -1.58
```

The output follows the familiar `htest` format. The reported p-value is the larger of the two one-sided tests against the equivalence bounds, and the confidence interval is the studentized bootstrap CI for the mean difference. Because the p-value is tied to this interval via CI inversion, a researcher reporting both will find the two tell a consistent story. The interval will fall within the equivalence bounds if and only if the equivalence test rejects. For situations where both a nil hypothesis test and an equivalence test are desired simultaneously with bootstrapping, `boot.t.TOST` returns a `TOSTt` object with bootstrapped SMD estimates (Kirby and Gerlanc 2013):

```
set.seed(891111)
test1 = boot.t.TOST(formula = extra ~ group,
                    data = sleep,
                    paired = TRUE,
                    eqb = .5,
                    R = 999)
print(test1)
```

```
##
## Bootstrapped Paired t-test
##
## The equivalence test was non-significant, t(9) = -2.777, p = 1e+00
## The null hypothesis test was significant, t(9) = -4.062, p = 0e+00
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test    -4.062  9 < 0.001
## TOST Lower -2.777  9      1
## TOST Upper -5.348  9 < 0.001
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw          -1.580 0.3509 [-2.9393, -1.0833]      0.9
## Hedges's g(z) -1.174 0.7990 [-1.3917, 1.1253]      0.9
## Note: studentized bootstrap ci method utilized.
```

The bootstrapped SMDs are a useful feature in their own right. Analytic confidence intervals for standardized effect sizes rely on approximations, which assume normality and can behave poorly with small or skewed samples. Bootstrap intervals for SMDs sidestep these assumptions by resampling the full estimation pipeline, yielding intervals that inherit the robustness properties of the bootstrap itself.

**6.2.1.1. Checking Assumptions.** The bootstrap t-test makes fewer distributional assumptions than its conventional, analytic, counterpart but is not assumption-

free. It requires independent observations and assumes the empirical distribution is a reasonable stand-in for the population distribution. The latter is an assumption that holds better in moderate-to-large samples than in small ones. The studentized bootstrap is reasonably robust to skewness and heavy tails, but remains sensitive to outliers and other high-influence observations, which can be drawn repeatedly across replicates and disproportionately shape the bootstrap distribution. Useful visual checks include the distribution of the data (histograms, boxplots) to identify extreme values, the bootstrap distribution itself to flag problematic skewness or gaps, and the jackknife-after-bootstrap plot to identify individual observations that dominate the bootstrap variability. The `tr` argument for trimmed means provides a straightforward robustness option when outliers are a concern. Diagnostic workflows for all three designs are demonstrated in the online supplement. As with the t-test, these checks reflect assumptions that are relaxed relative to normal-theory inference but not eliminated.

### 6.2.2. Permutation t-test

When `R = NULL`, the function enumerates all possible permutations for an exact test; otherwise, it draws `R` random permutations with the “+1” correction (Phipson and Smyth 2010). By default the test is studentized (`perm.se = TRUE`), consistent with the recommendations discussed in the methods section above. The `perm.se = FALSE` option is available primarily to match the behavior of other permutation test packages that do not studentize by default.

```
set.seed(8812)
perm.t.test(formula = extra ~ group,
             data = sleep,
             paired = TRUE,
             alternative = "equivalence",
             mu = c(-0.5, 0.5),
             R = 999)

##
## Randomization Permutation Paired t-test
##
## data: extra by group
## t-observed = -2.7766, df = 9, p-value = 0.999
## alternative hypothesis: equivalence
## null values:
## difference in means difference in means
## -0.5 0.5
## 90 percent confidence interval:
## -2.1720 -0.9956
## sample estimates:
## mean of the differences (z = '1' - '2')
## -1.58
```

Running the bootstrap and permutation equivalence tests on the same paired sleep data using matching equivalence bounds produces broadly similar conclusions, as would be expected. The two approaches can, however, diverge in small samples, under strong heteroscedasticity, or in the presence of influential outliers, which is where the choice between them becomes substantively important. Users interested in robustness

to outliers can additionally supply the `tr` argument to either function to use trimmed means, preserving the mean-based interpretation while downweighting extreme observations.

**6.2.2.1. *Checking Assumptions.*** The permutation t-test carries different assumptions depending on design. For the two-sample case, the studentized version is valid under exchangeability of observations across groups under the null and provides asymptotic validity even when variances differ between groups. For paired and one-sample designs, the sign-flip procedure additionally requires that the distribution of differences (or one-sample observations) is symmetric around the null value; asymmetry makes the test approximate rather than exact, though studentization provides some asymptotic protection. Symmetry is a commonly overlooked assumption for paired permutation tests and is easily checked with a symmetry plot of the differences, as demonstrated for the WMW signed-rank case above. For two-sample designs, overlaid empirical cumulative density functions and separate density plots help assess whether group shapes are comparable enough for exchangeability under the null to be credible. Diagnostic workflows for all three designs are demonstrated in the online supplement. As with the bootstrap, these checks reflect assumptions that are relaxed relative to Student's t-test but not eliminated.

### 6.3. Log-Transformed Methods

The natural log transformation stabilizes variance and approximates normality for right-skewed data (Bland and Altman 1996). An additional advantage is that differences on the log scale back-transform to *ratios*, making interpretation straightforward: if two samples have geometric means of 7 and 10.5, the ratio is 1.5 (i.e.,  $10.5/7$ ). This property is central to pharmaceutical bioequivalence, where the FDA requires comparisons on the log-transformed scale with equivalence bounds of 0.80 to 1.25 for the ratio of geometric means (Food and Drug Administration 2014; He et al. 2022).

An alternative to `log_TOST` is to log-transform data manually and analyze with `t.TOST`, `simple_hstest`, or the resampling methods. Differences on the log scale can then be interpreted as “sympercents” — symmetric percentage differences on the  $100 \cdot \ln(x) - \ln(y)$  scale (Cole 2000) — which are symmetric in a way that ordinary percentage differences are not because percentage changes are calculated from different bases, a 10% loss and a 10% gain do not cancel out, whereas +10 and -10 sympercents are exactly equal in magnitude.<sup>4</sup>

#### 6.3.1. Log TOST

The `log_TOST` function accepts paired and two-sample comparisons (one-sample tests are not supported since there is no ratio to calculate). Instead of an SMD, a ratio of means is reported (Lajeunesse 2015). The default equivalence bounds are set to the FDA standard (`eqb` = 1.25; the reciprocal is taken as the other bound).

```
log_TOST(mpg ~ am, data = mtcars)

##
## Log-transformed Welch Two Sample t-test
##
## The equivalence test was non-significant, t(23.96) = -1.363, p = 9.07e-01
## The null hypothesis test was significant, t(23.96) = -3.826, p = 8.19e-04
## NHST: reject null significance hypothesis that the effect is equal to one
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t      df p.value
## t-test    -3.826 23.96 < 0.001
## TOST Lower -1.363 23.96  0.907
## TOST Upper -6.288 23.96 < 0.001
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## log(Means Ratio) -0.3466 0.09061 [-0.5017, -0.1916]      0.9
## Means Ratio      0.7071      NA  [0.6055, 0.8256]      0.9
```

---

<sup>4</sup>A concrete illustration that I give my students: if a stock falls 10% on Tuesday and rises 10% from Tuesday on Wednesday, you still have a loss on of your original investment. Sympercents do not have this asymmetry.

### 6.3.2. Bootstrap Log TOST

The bootstrap version uses the same bootstrap methods as `boot_t_test` on the log-transformed values:

```
boot_log_TOST(mpg ~ am, data = mtcars, R=999)

##
## Bootstrapped Log Welch Two Sample t-test
##
## The equivalence test was non-significant, t(23.96) = -1.363, p = 9.54e-01
## The null hypothesis test was significant, t(23.96) = -3.826, p = 0e+00
## NHST: reject null significance hypothesis that the effect is equal to 1
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t      df p.value
## t-test    -3.826 23.96 < 0.001
## TOST Lower -1.363 23.96  0.954
## TOST Upper -6.288 23.96 < 0.001
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## log(Means Ratio) -0.3466 0.08621 [-0.4932, -0.1893]      0.9
## Means Ratio      0.7071 0.06116 [0.6106, 0.8275]      0.9
## Note: studentized bootstrap ci method utilized.
```

From this analysis, we would conclude there is a significant effect that is not practically equivalent.

### 6.3.3. Checking Assumptions

The log-transformed methods inherit the assumptions of the underlying test applied to the transformed data. The `log_TOST` function carries forward the t-test assumptions, and `boot_log_TOST` carries forward the bootstrap t-test assumptions. What changes is where those assumptions need to hold. Normality, symmetry, and homogeneity of variance should be assessed on the log scale rather than the original scale, since that is where inference takes place. The same visual diagnostics apply just computed on log-transformed data. This points to a broader practical use of log transformation. When the symmetry assumption of the sign-flip permutation test is violated because paired differences are right-skewed on the original scale, the differences of logs are often approximately symmetric, and `perm_t_test` on log-transformed values becomes a reasonable option. In such cases the equivalence bounds should be specified on the log scale and interpreted as log ratios (or sympercents). As with Student's t-test or the bootstrap t-test, assumptions are relaxed by the transformation but not eliminated — they are simply moved to a different scale.



## 7. Equivalence Testing with ANOVAs

Many researchers use ANOVA as an omnibus test before inspecting pairwise comparisons, particularly in factorial designs. As [Campbell and Lakens \(2021\)](#) note, the lack of a significant ANOVA result does not indicate that a factor has no effect. Equivalence testing for ANOVAs addresses this directly by asking: can we reject the hypothesis that the partial  $\eta^2$  exceeds a specified bound?

The equivalence test uses the non-central  $F$  distribution. The non-centrality parameter is computed from the equivalence bound and the degrees of freedom, and the  $p$ -value is obtained from the cumulative distribution function of the non-central  $F$ -value. I have extended the approach of [Campbell and Lakens \(2021\)](#) — who proposed equivalence testing for one-way ANOVAs — to accommodate multi-factor and factorial designs by generalizing the non-centrality parameter calculation. The full derivation is described in the package documentation; here we demonstrate usage.

The equivalence test inherits the standard ANOVA assumptions which should be checked on the fitted model itself rather than on the equivalence output. The `performance::check_model()` function provides a comprehensive visual diagnostic suite for most linear model objects in R on a single call; the online supplement demonstrates its use.

### 7.1. Example with Summary Statistics

Using the `InsectSprays` data and the `equ_ftest` function, which accepts summary  $F$ -test statistics directly:

```
data("InsectSprays")
aovtest = aov(count ~ spray, data = InsectSprays)
anova(aovtest)

## Analysis of Variance Table
##
## Response: count
##           Df Sum Sq Mean Sq F value    Pr(>F)
## spray      5 2668.8   533.77   34.702 < 2.2e-16 ***
## Residuals 66 1015.2    15.38
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

equ_ftest(Fstat = 34.70228, df1 = 5, df2 = 66, eqb = 0.35)

##
## Equivalence Test from F-test
##
## data: Summary Statistics
## F = 34.702, df1 = 5, df2 = 66, p-value = 1
## 95 percent confidence interval:
##  0.5806263 0.7804439
## sample estimates:
## [1] 0.724439
```

Based on these results, there is a significant effect of “spray” and the differences are *not* statistically equivalent — we reject the traditional null hypothesis but fail to reject the equivalence null hypothesis.

## 7.2. Example with R ANOVA Objects

The `equ_anova` function accepts objects from `stats::aov`, `car::Anova`, and `afex::aov_car`, avoiding the need to extract summary statistics manually:

```
# Example using a purely within-subjects design
# (Maxwell & Delaney, 2004, Chapter 12, Table 12.5, p. 578):
library(afex)
data(md_12.1)
aovtest2 = aov_ez("id", "rt", md_12.1, within = c("angle", "noise"),
  anova_table=list(correction = "none", es = "none"))
equ_anova(aovtest2,
  eqb = 0.35)
```

| ##   | effect      | df1 | df2 | F.value   | p.null       | pes       | eqbound | p.equ     |
|------|-------------|-----|-----|-----------|--------------|-----------|---------|-----------|
| ## 1 | (Intercept) | 1   | 9   | 598.44917 | 1.526600e-09 | 0.9851839 | 0.35    | 0.9999997 |
| ## 2 | angle       | 2   | 18  | 40.71910  | 2.086763e-07 | 0.8189831 | 0.35    | 0.9992557 |
| ## 3 | noise       | 1   | 9   | 33.76596  | 2.559737e-04 | 0.7895522 | 0.35    | 0.9763228 |
| ## 4 | angle:noise | 2   | 18  | 45.31034  | 9.424093e-08 | 0.8342857 | 0.35    | 0.9996103 |

## 8. Power Analysis and Sample Size Planning

Researchers planning an equivalence study will need to determine the required sample size. Power for equivalence tests behaves differently than for standard tests: larger samples are generally needed, and power depends on both the true effect size and the width of the equivalence bounds.

### 8.1. Power for TOST *t*-tests

The `power.t.TOST` function computes exact power for TOST with one-sample, two-sample, and paired *t*-tests. It can solve for any one of `n`, `power`, or `alpha` given the other parameters are provided. The `delta` argument specifies the true mean difference (set to 0 if the true effect is exactly zero) and `eqb` specifies the equivalence bounds.

For example, to determine the sample size needed to achieve 80% power to detect equivalence within bounds of  $\pm 0.5$  (assuming the true difference is zero and  $SD = 1$ ):

```
power.t.TOST(delta = 0, sd = 1, eqb = 0.5,
             alpha = 0.05, power = 0.8,
             type = "two.sample")
```

```
##
##      Two-sample TOST power calculation
##
##           power = 0.8
##           beta  = 0.2
##           alpha = 0.05
##           n     = 69.19782
##           delta = 0
##           sd    = 1
##           bounds = -0.5, 0.5
##
## NOTE: n is number in *each* group
```

### 8.2. Power for Correlations

The `power.z.cor` function provides power analysis for equivalence tests on Pearson correlations using Fisher's *z*-transformation:

```
power.z.cor(rho = 0, power = 0.8,
            null = 0.3, alpha = 0.05,
            alternative = "equivalence")
```

```
##
##      Approximate Power for Pearson Product-Moment Correlation (z-test)
##
##           n = 92.38991
##           rho = 0
##           alpha = 0.05
```

```
##          beta = 0.2
##          power = 0.8
##          null = 0.3, -0.3
##          alternative = equivalence
```

### 8.3. *Extensions: PowerTOST and Superpower*

For more complex designs — crossover studies, replicate designs, group-sequential equivalence trials — the PowerTOST package ([Labes, Schütz, and Lang 2021](#)) provides extensive functionality beyond what TOSTER offers. For factorial designs where equivalence contrasts are tested via estimated marginal means, simulation-based power analysis via the Superpower package is recommended ([Lakens and Caldwell 2021](#)). Both packages complement TOSTER's built-in power tools for common research designs.

## 9. Conclusions

This paper has described the major updates to the TOSTER R package, expanding its scope from the original t-test and correlation equivalence tests to a comprehensive toolkit for frequentist equivalence testing. The update adds a unified t-test interface with minimal effect testing support, a simplified hypothesis testing interface that integrates with the standard R ecosystem, tools for standardized effect sizes and correlations, robust and resampling-based alternatives, ANOVA equivalence testing, and power analysis for sample size planning.

Several limitations should be acknowledged. TOSTER only covers a few standard research designs — it does not handle many analysis scenarios such as those involving multilevel or mixed-effects models. For these settings, researchers should consider packages like “emmeans” or “marginaleffects”, which can accommodate non-zero null hypotheses in more complex modeling frameworks. TOSTER also does not provide Bayesian equivalence testing; researchers interested in Bayesian methods for the null should consult other R packages such as “bayestestR”.

I hope that the updated package continues to build upon the original impact of [Lakens \(2017\)](#) and makes equivalence testing more accessible to researchers across disciplines. At the very least, I would encourage users to compare results across methods (e.g., Student’s t-test vs. permutation t-test vs. bootstrap t-test) to ensure that conclusions do not entirely depend on the chosen analysis. If the interpretation substantially changes depending on the test, this suggests features of the data — violations of assumptions, outliers, distributional shape — that warrant further investigation.

## 10. Additional Information

The TOSTER package can be installed from CRAN or from GitHub:

```
# Install from CRAN
install.packages("TOSTER")

# Install the development version from GitHub
devtools::install_github("Lakens/TOSTER")
```

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### *Disclosure statement*

The author of this manuscript is the author of the TOSTER package. Citations of this manuscript will benefit his citation count.

### *AI Use Disclosure*

During the preparation of the TOSTER package updates and this manuscript, the author used Claude Opus 4.5-4.7 (Anthropic) to assist with editing R code and drafting unit tests. For the manuscript text, Claude Opus was used only for targeted copy-editing — suggesting revisions to specific passages, catching typographical and grammatical errors, and identifying inconsistencies across sections. The manuscript was written by the author, and all AI-assisted output was reviewed, revised, and verified by the author, who takes full responsibility for the content of the package and the manuscript.

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### *Nomenclature/Notation*

- ANOVA: Analysis of Variance
- Bootstrapping: the use of random sampling with replacement to estimate statistics
- FDA: Food and Drug Administration (United States of America)
- MET: Minimal Effects Test

- ncp: non-centrality parameter
- SESOI: Smallest Effect Size of Interest
- SMD: Standardized Mean Difference (e.g., Cohen's d)
- TOST: Two One-Sided Tests
- WMW: Wilcoxon-Mann-Whitney

### *Notes*

The R package is also (partially) implemented in [jamovi](#) as the TOSTER module.



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